

COMPOSITE VS. STEEL ROLLER: 10-POINT COMPARISON

A side-by-side look at how composite conveyor rollers stack up against standard steel idlers, based on published field data and product specifications.

FACTOR	COMPOSITE ROLLER	STANDARD STEEL IDLER
01 WEIGHT	50-60% lighter than an equivalent steel roller, depending on troughing vs. return application.	Baseline weight. Heavier shell adds up quickly across long conveyor flights.
02 OPERATING NOISE	≥70% quieter (minimum, per published field data) than steel rollers in operation.	Steel-on-steel contact and shell resonance typically produce the loudest running idler on the belt line.
03 POWER CONSUMPTION / ROLLING FRICTION	8-26% lower running friction / power draw vs. steel (model-dependent — see manufacturer spec sheet).	Baseline friction and startup draw. Higher rotating mass increases motor startup load.
04 CORROSION RESISTANCE	HDPE / UHMWPE shell does not rust. Stainless steel shaft and bearing options available for corrosive environments.	Prone to rust and corrosion over time, especially in humid, coastal, or wash-down environments, without protective coating.
05 ABRASION RESISTANCE	UHMWPE tube rated up to 3.4x more abrasion-resistant than steel in published mass-loss comparisons.	Baseline abrasion resistance. Wears faster under high-abrasion material contact without a hardened liner.
06 IMPACT RESISTANCE	UHMWPE and dual-layer lines rated for "Ultimate" impact resistance; dual-layer construction includes a visible wear-indicator outer skin.	Solid steel shell resists impact but dents and deforms under heavy point-loading; no wear-indicator layer.
07 FIRE RETARDANT / ANTI-STATIC OPTIONS	FR and FRAS variants self-extinguish in under 6 seconds (tested to MSHA IC 335/01, V0 rating); anti-static FRAS option available (<300,000 Ω).	Steel itself is non-combustible; FR/anti-static coatings or linings, where required, are a separate added spec.
08 CHEMICAL / ACID RESISTANCE	Rated "Excellent" acid resistance across composite product lines.	Susceptible to acid and chemical corrosion without specialized coating or stainless construction.
09 INSTALLATION & HANDLING	Lighter roller sets ease manual handling during installation and replacement, reducing lifting strain and time per swap.	Heavier units typically require more manual effort, more hands, or lifting aids per roller change.
10 WARRANTY	12-month standard warranty on entry-level composite lines (CCR Base Grey).	Varies by manufacturer and product line.

WANT THE NUMBERS FOR YOUR SPECIFIC CONVEYOR LINE?

Our engineers can run a formal site assessment and quote.

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Figures reflect Composite Roller Co.'s published performance data (compositeroller.com and product spec sheets) and are directional, not guaranteed for every application. Actual performance varies by belt speed, load, environment, and roller specification. Steel idler characteristics reflect typical, uncoated CEMA-standard construction and may vary by manufacturer. Contact our engineers for a formal assessment specific to your site.